

SUPPLEMENTAL MATERIAL

Table S1. Univariate and Multivariate Conditional Logistic Regression: Small LV.

Variable	Univariate analysis		Multivariate analysis	
	Odds ratio (95% CI)	P value	Odds ratio (95% CI) ^a	P value
Age, per SD increase	1.68 (1.63-1.74)	<0.001	1.81 (1.73-1.88)	<0.001
Females	1.08 (1.01-1.15)	0.02	2.04 (1.85-2.25)	<0.001
Non-whites	1.83 (1.57-2.15)	<0.001	2.25 (1.90-2.68)	<0.001
Less education	1.27 (1.19-1.35)	<0.001	1.13 (1.05-1.21)	0.001
Never drinking	1.61 (1.44-1.78)	<0.001	1.34 (1.11-1.61)	0.002
Current smoking	1.50 (1.33-1.69)	<0.001	1.41 (1.23-1.60)	<0.001
Clinical history, %				
Coronary heart disease	1.29 (1.14-1.47)	<0.001	1.05 (0.88-1.25)	0.20
Stroke	1.45 (1.11-1.91)	0.007	1.05 (0.79-1.40)	0.74
Heart failure	1.68 (1.39-2.04)	<0.001	1.19 (0.97-1.47)	0.09
Hypertension	1.76 (1.63-1.89)	<0.001	1.22 (1.13-1.33)	<0.001
Diabetes	2.81(2.43-3.25)	<0.001	1.88 (1.56-2.28)	<0.001
Townsend deprivation index, per SD increase	1.07 (1.03-1.10)	<0.001	1.07 (1.04-1.11)	<0.001
Physical activity, per SD decrease	1.12 (1.08-1.16)	<0.001	1.08 (1.03-1.13)	0.001
Systolic blood pressure, per SD increase	1.17 (1.13-1.21)	<0.001	1.45 (1.35-1.55)	<0.001
Diastolic blood pressure, per SD increase	1.24 (1.20-1.28)	<0.001	1.48 (1.39-1.58)	<0.001
Body mass index, per SD increase	1.31 (1.27-1.34)	<0.001	1.05 (1.01-1.09)	0.009
Laboratory findings				
Hb, per SD increase	1.24 (1.20-1.28)	<0.001	1.33 (1.27-1.40)	<0.001
WBC, per SD increase	1.21 (1.17-1.25)	<0.001	1.06 (1.03-1.10)	<0.001
Albumin, per SD decrease	1.08 (1.04-1.11)	<0.001	1.03 (0.99-1.07)	0.12
C-reactive protein, per SD increase	1.14 (1.11-1.16)	<0.001	1.06 (1.02-1.09)	0.001
Fasting glucose, per SD increase	1.13 (1.10-1.17)	<0.001	0.98 (0.95-1.02)	0.41

HbA1c, per SD increase	1.30 (1.27-1.34)	<0.001	1.09 (1.04-1.13)	<0.001
Total cholesterol, per SD increase	1.07 (1.03-1.11)	0.001	1.03 (0.92-1.15)	0.61
LDL-cholesterol, per SD increase	1.18 (1.14-1.22)	<0.001	1.05 (0.96-1.16)	0.30
HDL-cholesterol, per SD decrease	1.26 (1.21-1.32)	<0.001	1.09 (1.01-1.18)	0.04
Triglycerides, per SD increase	1.35 (1.31-1.38)	<0.001	1.21 (1.17-1.26)	<0.001

Hb, hemoglobin concentration; WBC, white blood cell count; HbA1c, hemoglobin A1c; LDL, low-density lipoprotein; HDL, high-density lipoprotein

Table S2. The Association Between LV Size and Other Outcomes.

Outcomes	Events/Person-y	Incidence Rate (95%CI), per 1,000 Person-y	Risk difference(95%CI), per 1,000 Person-y	Model 1	Model 2	Model 3
Atrial fibrillation						
Normal LV	765/195535	3.9 (3.6-4.2)	Ref.	Ref.	Ref.	Ref.
Small LV	114/24110	4.7 (3.9-5.7)	0.8 (0-1.7)	1.25 (1.02-1.51)	1.08 (0.89-1.32)	1.06 (0.87-1.30)
Large LV	60/5916	10.1 (7.9-13.1)	6.2 (3.6-8.8)	2.56 (1.97-3.33)	2.41 (1.85-3.14)	2.42 (1.86-3.15)
Hemorrhagic stroke						
Normal LV	49/200268	0.2 (0.1-0.3)	Ref.	Ref.	Ref.	Ref.
Small LV	6/24642	0.2 (0.1-0.5)	0 (-0.2-0.2)	1.02 (0.44-2.39)	0.90 (0.38-2.13)	0.85 (0.36-2.02)
Large LV	3/6287	0.5 (0.2-1.5)	0.2 (0-0.8)	1.94 (0.61-6.23)	2.00 (0.62-6.45)	2.02 (0.63-6.51)
Cardiovascular death						
Normal LV	110/193527	0.6 (0.5-0.7)	Ref.	Ref.	Ref.	Ref.
Small LV	14/23961	0.6 (0.3-1.0)	0 (-0.3-0.3)	1.07 (0.62-1.87)	0.83 (0.47-1.46)	0.80 (0.45-1.41)
Large LV	11/6114	1.8 (1.0-3.2)	1.2 (0.2-2.3)	3.03 (1.63-5.63)	2.90 (1.55-5.41)	2.95 (1.58-5.51)
Noncardiovascular death						
Normal LV	476/193527	2.5 (2.2-2.7)	Ref.	Ref.	Ref.	Ref.
Small LV	95/23961	4.1 (3.4-5.0)	1.5 (0.7-2.3)	1.67 (1.34-2.08)	1.52 (1.22-1.90)	1.47 (1.17-1.84)
Large LV	16/6114	3.2 (2.1-4.9)	0.2 (-1.1-1.5)	1.03 (0.63-1.70)	0.95 (0.58-1.57)	0.98 (0.59-1.610)

Model 1: Adjusted for age, sex and race.

Model 2: Adjusted for age, sex, race, education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index and total cholesterol.

Model 3: Adjusted for age, sex, race, education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and LVEF.

Table S3. The Association Between LV Size and Cardiovascular Outcomes Using UK biobank Cutoff Values.

Outcomes	Events/Person-y	Incidence Rate (95%CI),per 1,000 Person-y	Risk difference(95%CI), per 1,000 Person-y	Model 1	Model 2	Model 3
MACE						
Normal LV	1240/178519	6.9 (6.6-7.3)	Ref.	Ref.	Ref.	Ref.
Small LV	193/17796	10.8 (9.4-12.5)	3.7 (2.2-5.3)	1.51 (1.29-1.75)	1.28 (1.10-1.50)	1.26 (1.07-1.48)
Large LV	67/5515	12.1 (9.6-15.4)	5.1 (2.3-8.0)	1.84 (1.44-2.35)	1.80 (1.40-2.30)	1.77 (1.36-2.29)
Coronary heart disease						
Normal LV	909/181975	5.0 (4.7-5.3)	Ref.	Ref.	Ref.	Ref.
Small LV	150/18388	8.1 (7.0-9.6)	3.2 (1.8-4.5)	1.56 (1.31-1.85)	1.32 (1.11-1.57)	1.29 (1.07-1.55)
Large LV	37/5729	6.5 (4.7-8.9)	1.5 (-0.6-3.6)	1.36 (0.98-1.89)	1.33 (0.96-1.85)	1.44 (1.03-2.01)
Heart failure						
Normal LV	296/204508	1.4 (1.3-1.6)	Ref.	Ref.	Ref.	Ref.
Small LV	34/21216	1.6 (1.1-2.2)	0.2 (-0.4-0.7)	1.05 (0.74-1.50)	0.84 (0.59-1.20)	0.76 (0.52-1.12)
Large LV	62/6488	9.6 (7.5-12.3)	8.1 (5.7-10.5)	6.94 (5.28-9.13)	6.63 (5.03-8.74)	6.83 (5.10-9.15)
Ischemic stroke						
Normal LV	231/204805	1.1 (1.0-1.3)	Ref.	Ref.	Ref.	Ref.
Small LV	43/21220	2.0 (1.5-2.7)	0.9 (0.3-1.5)	1.75 (1.26-2.42)	1.52 (1.09-2.11)	1.48 (1.04-2.10)
Large LV	14/6782	2.1 (1.2-3.5)	0.9 (-0.1-2.0)	1.87 (1.09-3.21)	1.72 (1.01-2.96)	1.45 (0.79-2.67)
Death from any cause						
Normal LV	593/196601	3.0 (2.8-3.3)	Ref.	Ref.	Ref.	Ref.
Small LV	99/20467	4.8 (4.0-5.9)	1.8 (0.8-2.8)	1.57 (1.27-1.94)	1.38 (1.11-1.71)	1.36 (1.09-1.71)
Large LV	30/6534	4.6 (3.2-6.6)	1.6 (0-3.2)	1.55 (1.08-2.24)	1.43 (0.99-2.07)	1.33 (0.89-1.99)

Model 1: Adjusted for age, sex and race.

Model 2: Adjusted for age, sex, race, education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity,

systolic blood pressure, diastolic blood pressure, body mass index and total cholesterol.

Model 3: Adjusted for age, sex, race, education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and LVEF.

Table S4. The Association Between LV Size and Cardiovascular Outcomes in Competitive Risk Regression Analysis.

Outcomes	Events/Person-y	Incidence Rate (95%CI), per 1,000 Person-y	Adjusted subdistribution HR (95%CI)*
MACE			
Normal LV	1240/178519	6.9 (6.6-7.3)	Ref.
Small LV	193/17796	10.8 (9.4-12.5)	1.34 (1.15-1.56)
Large LV	67/5515	12.1 (9.6-15.4)	1.72 (1.32-2.23)
Coronary heart disease			
Normal LV	909/181975	5.0 (4.7-5.3)	Ref.
Small LV	150/18388	8.1 (7.0-9.6)	1.40 (1.17-1.66)
Large LV	37/5729	6.5 (4.7-8.9)	1.38 (1.01-1.90)
Heart failure			
Normal LV	296/204508	1.4 (1.3-1.6)	Ref.
Small LV	34/21216	1.6 (1.1-2.2)	0.88 (0.62-1.25)
Large LV	62/6488	9.6 (7.5-12.3)	6.54 (4.91-8.72)
Ischemic stroke			
Normal LV	231/204805	1.1 (1.0-1.3)	Ref.
Small LV	43/21220	2.0 (1.5-2.7)	1.57 (1.13-2.17)
Large LV	14/6782	2.1 (1.2-3.5)	1.69 (0.96-2.98)

*Adjusted for age, sex, race, education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and LVEF.

Table S5. The association Between LV Size and Cardiovascular Outcomes by Excluding Subjects with Baseline Cardiovascular Disease.

Outcomes	Events/Person-y	Incidence Rate (95%CI), per 1,000 Person-y	Risk difference(95%CI), per 1,000 Person-y	Model 1	Model 2	Model 3
MACE						
Normal LV	1174/173979	6.7 (6.4-7.1)	Ref.	Ref.	Ref.	Ref.
Small LV	200/20494	9.8 (8.5-11.2)	3.0 (1.6-4.4)	1.49 (1.28-1.73)	1.28 (1.10-1.49)	1.23 (1.03-1.46)
Large LV	36/4171	8.6 (6.2-12.0)	1.9 (-1.0-4.7)	1.25 (0.90-1.74)	1.26 (0.90-1.76)	1.25 (0.85-1.84)
Coronary heart disease						
Normal LV	849/175428	4.8 (4.5-5.2)	Ref.	Ref.	Ref.	Ref.
Small LV	151/20817	7.3 (6.2-8.5)	2.4 (1.2-3.6)	1.54 (1.29-1.83)	1.32 (1.10-1.57)	1.24 (1.04-1.48)
Large LV	22/4222	5.2 (3.4-7.9)	0.4 (-0.2-2.6)	1.05 (0.69-1.60)	1.07 (0.70-1.64)	1.22 (0.77-1.93)
Heart failure						
Normal LV	193/187856	1.0 (0.9-1.2)	Ref.	Ref.	Ref.	Ref.
Small LV	26/22433	1.2 (0.8-1.7)	0.1 (-0.3-0.6)	1.16 (0.77-1.74)	0.93 (0.61-1.40)	1.02 (0.65-1.59)
Large LV	14/4499	3.1 (1.8-5.3)	2.1 (0.4-3.7)	2.97 (1.72-5.10)	3.12 (1.80-5.39)	3.71 (2.04-6.74)
Ischemic stroke						
Normal LV	190/187796	1.0 (0.9-1.2)	Ref.	Ref.	Ref.	Ref.
Small LV	38/22402	1.7 (1.2-2.3)	0.7 (0.1-1.2)	1.70 (1.20-2.41)	1.49 (1.05-2.13)	1.60 (1.07-2.37)
Large LV	5/4527	1.1 (0.5-2.7)	0.1 (-0.8-1.1)	1.08 (0.44-2.63)	1.03 (0.42-2.51)	0.84 (0.27-2.66)
Death from any cause						
Normal LV	474/179549	2.6 (2.4-2.9)	Ref.	Ref.	Ref.	Ref.
Small LV	93/21435	4.3 (3.5-5.3)	1.7 (0.8-2.6)	1.71 (1.37-2.13)	1.51 (1.21-1.90)	1.40 (1.08-1.81)
Large LV	9/4333	2.1 (1.1-4.0)	-0.6 (-1.9-0.8)	0.76 (0.39-1.48)	0.75 (0.39-1.46)	0.43 (0.16-1.15)

Model 1: Adjusted for age, sex and race.

Model 2: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity,

systolic blood pressure, diastolic blood pressure, body mass index and total cholesterol.

Model 3: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and LVEF.

Table S6. Baseline Characteristics according to LV size and Sex Category.

Factor	Men				Women			
	Normal LV (n=16277)	Small LV (n=2003)	Large LV (n=556)	p-value	Normal LV (n=17714)	Small LV (n=2302)	Large LV (n=511)	p-value
Age (years)	64.0 (7.7)	67.3 (6.9)	63.3 (8.2)	<0.001	62.6 (7.4)	66.6 (6.7)	61.5 (7.7)	<0.001
Whites,%	15832 (97.3)	1886 (94.2)	543 (97.7)	<0.001	17288 (97.6)	2221 (96.5)	488 (95.5)	<0.001
Higher education,%	7858 (48.3)	852 (42.5)	276 (49.6)	<0.001	7898 (44.6)	892 (38.7)	245 (47.9)	<0.001
Current drinking status,%	15655 (96.2)	1904 (95.1)	542 (97.5)	0.012	16787 (94.8)	2119 (92.1)	485 (94.9)	<0.001
Current smoking status,%	1141 (7.0)	205 (10.2)	44 (7.9)	<0.001	888 (5.0)	142 (6.2)	38 (7.4)	0.004
Clinical history, %								
Coronary heart disease	1314 (8.1)	186 (9.3)	91 (16.4)	<0.001	502 (2.8)	107 (4.6)	35 (6.8)	<0.001
Stroke	216 (1.3)	41 (2.0)	14 (2.5)	0.004	122 (0.7)	21 (0.9)	2 (0.4)	0.33
Heart failure	456 (2.8)	87 (4.3)	137 (24.6)	<0.001	152 (0.9)	41 (1.8)	109 (21.3)	<0.001
Hypertension	5518 (33.9)	980 (48.9)	190 (34.2)	<0.001	4016 (22.7)	769 (33.4)	127 (24.9)	<0.001
Diabetes	1007 (6.2)	292 (14.6)	35 (6.3)	<0.001	534 (3.0)	204 (8.9)	13 (2.5)	<0.001
Townsend deprivation index	-2.0 (2.7)	-1.7 (2.9)	-2.0 (2.6)	<0.001	-1.9 (2.7)	-1.8 (2.7)	-1.7 (2.7)	0.17
Physical activity, met-min/week	2500.2 (2457.4)	2248.5 (2411.3)	3016.0 (2685.7)	<0.001	2430.9 (2313.2)	2184.3 (2228.5)	2774.3 (2320.6)	<0.001
Systolic blood pressure, mmHg	136.2 (20.6)	138.7 (21.1)	136.0 (23.6)	<0.001	128.5 (21.7)	132.6 (22.1)	129.0 (22.9)	<0.001
Diastolic blood pressure, mmHg	81.9 (12.2)	84.1 (12.5)	79.2 (13.8)	<0.001	77.6 (12.1)	80.4 (12.3)	75.4 (12.7)	<0.001
Body mass index,kg/m2	26.9 (3.6)	28.1 (4.2)	26.7 (3.7)	<0.001	25.9 (4.5)	27.2 (4.6)	25.6 (4.6)	<0.001
Cardiovascular medications, %								

Antihypertensive	4052 (24.9)	732 (36.5)	165 (29.7)	<0.001	2702 (15.3)	516 (22.4)	77 (15.1)	<0.001
Antiplatelet	1903 (11.7)	295 (14.7)	79 (14.2)	<0.001	795 (4.5)	147 (6.4)	31 (6.1)	<0.001
Lipid-lowering	4162 (25.6)	743 (37.1)	137 (24.6)	<0.001	1961 (11.1)	510 (22.2)	46 (9.0)	<0.001
Beta blocker	1041 (6.4)	143 (7.1)	72 (12.9)	<0.001	641 (3.6)	74 (3.2)	21 (4.1)	0.50
Diuretics	489 (3.0)	111 (5.5)	23 (4.1)	<0.001	557 (3.1)	130 (5.7)	18 (3.5)	<0.001
Laboratory findings								
Hb, g/dL	15.0 (0.9)	15.2 (1.0)	14.7 (0.9)	<0.001	13.4 (0.9)	13.7 (0.9)	13.2 (1.0)	<0.001
WBC, $\times 10^3/\text{mm}^3$	6.5 (1.9)	7.0 (1.8)	6.2 (1.6)	<0.001	6.6 (1.7)	6.8 (1.7)	6.3 (1.6)	<0.001
Albumin, g/dL	45.7 (2.5)	45.5 (2.5)	45.9 (2.7)	<0.001	45.1 (2.5)	45.0 (2.5)	45.0 (2.5)	0.091
C-reactive protein, g/dL	1.9 (3.4)	2.5 (3.9)	1.7 (3.0)	<0.001	2.0 (3.5)	2.7 (3.6)	1.8 (4.8)	<0.001
Fasting glucose, mean (SD)	5.0 (1.0)	5.2 (1.3)	5.0 (0.8)	<0.001	4.9 (0.8)	5.1 (1.2)	4.9 (0.7)	<0.001
HbA1c, mean (SD)	35.1 (5.4)	36.8 (7.0)	35.0 (4.9)	<0.001	34.6 (4.4)	36.4 (6.3)	34.1 (4.4)	<0.001
Total cholesterol, mmol/L	4.5 (0.9)	4.4 (0.9)	4.4 (0.8)	<0.001	4.8 (0.9)	5.0 (0.9)	4.7 (0.9)	<0.001
LDL-cholesterol, mmol/L	3.6 (0.8)	3.6 (0.9)	3.4 (0.8)	<0.001	3.6 (0.8)	3.8 (0.9)	3.4 (0.8)	<0.001
HDL-cholesterol, mmol/L	1.2 (0.3)	1.1 (0.2)	1.3 (0.3)	<0.001	1.5 (0.3)	1.4 (0.3)	1.5 (0.3)	<0.001
Triglycerides, mmol/L	1.9 (1.1)	2.2 (1.2)	1.6 (0.9)	<0.001	1.4 (0.7)	1.8 (0.9)	1.2 (0.7)	<0.001
CMR findings								
LVEDV, ml	158.7 (25.6)	108.0 (14.2)	248.8 (57.7)	<0.001	123.6 (19.1)	88.3 (11.1)	208.5 (82.9)	<0.001
LVEDVi, mL/m ²	79.5 (11.3)	53.5 (5.4)	125.1 (25.4)	<0.001	71.3 (9.0)	50.7 (4.4)	120.6 (46.0)	<0.001
LVESV, ml	72.6 (16.4)	49.3 (10.2)	135.6 (60.9)	<0.001	53.3 (11.2)	37.7 (7.5)	114.8 (76.3)	<0.001
LVESVi, mL/m ²	36.3 (7.6)	24.4 (4.5)	68.0 (28.7)	<0.001	30.8 (5.9)	21.7 (3.8)	66.3 (43.3)	<0.001
LVSVMl	86.1 (16.1)	58.7 (11.0)	113.3 (26.2)	<0.001	70.3 (12.4)	50.5 (8.6)	93.7 (23.2)	<0.001
LVSVi, mL/m ²	43.1 (7.5)	29.1 (5.0)	57.1 (12.9)	<0.001	40.6 (6.1)	29.0 (4.2)	54.3 (12.6)	<0.001
LVEF, %	54.4 (6.2)	54.4 (7.3)	46.9 (11.3)	<0.001	56.9 (5.5)	57.2 (6.6)	48.2 (12.1)	<0.001

Expressed as mean and SD or percentages.

There is statistically significant difference ($P < 0.05$) between normal left ventricle (LV) and small LV groups.

Hb, hemoglobin concentration; WBC, white blood cell count; HbA1c, hemoglobin A1c; LDL, low-density lipoprotein; HDL, high-density lipoprotein; CMR, cardiovascular magnetic resonance; LVEDV, left ventricular end-diastolic volume; LVEDVi, left ventricular end-diastolic volume index; LVESV, left ventricular end-systolic volume; LVESVi, left ventricular end-systolic volume index; LVSV, left ventricular stroke volume; LVSVi, left ventricular stroke volume index; LVEF, left ventricular ejection fraction.

Table S7. The Association Between LV Size and Cardiovascular Outcomes Stratified by Sex.

Outcomes	Events/Person-y	Incidence Rate (95%CI),per 1,000 Person-y	Risk difference(95%CI), per 1,000 Person-y	Model 1	Model 2	Model 3
Men						
MACE						
Normal LV	767/80638	9.5 (8.9-10.2)	Ref.	Ref.	Ref.	Ref.
Small LV	144/9219	15.6 (13.3-18.4)	6.1 (3.5-8.7)	1.66 (1.39-1.98)	1.42 (1.18-1.70)	1.35 (1.10-1.65)
Large LV	49/2404	20.4 (15.4-27.0)	10.9 (5.1-16.6)	2.14 (1.61-2.86)	2.11 (1.58-2.82)	2.10 (1.51-2.91)
Coronary heart disease						
Normal LV	589/82700	7.1 (6.6-7.7)	Ref.	Ref.	Ref.	Ref.
Small LV	113/9649	11.7 (9.7-14.1)	4.6 (2.4-6.8)	1.65 (1.35-2.02)	1.41 (1.15-1.73)	1.30 (1.03-1.64)
Large LV	30/2560	11.7 (8.2-16.8)	4.6 (0.4-8.8)	1.65 (1.14-2.38)	1.62 (1.12-2.34)	1.79 (1.21-2.67)
Heart failure						
Normal LV	195/95641	2.0 (1.8-2.3)	Ref.	Ref.	Ref.	Ref.
Small LV	25/11442	2.2 (1.5-3.2)	0.1 (-0.7-1.0)	1.08 (0.71-1.64)	0.85 (0.56-1.30)	0.81 (0.51-1.29)
Large LV	46/3007	15.3 (11.5-20.4)	13.3 (8.8-17.7)	7.53 (5.46-10.38)	7.12 (5.15-9.86)	8.10 (5.67-11.58)
Ischemic stroke						
Normal LV	129/95977	1.3 (1.1-1.6)	Ref.	Ref.	Ref.	Ref.
Small LV	32/11434	2.8 (2.0-4.0)	1.5 (0.5-2.5)	2.09 (1.42-3.09)	1.87 (1.26-2.77)	1.62 (1.02-2.56)
Large LV	11/3228	3.5 (2.0-6.2)	2.1 (0.1-4.1)	2.53 (1.37-4.69)	2.35 (1.27-4.37)	1.58 (0.69-3.61)
Death from any cause						
Normal LV	376/92370	4.1 (3.7-4.5)	Ref.	Ref.	Ref.	Ref.
Small LV	75/11084	6.8 (5.4-8.5)	2.7 (1.1-4.3)	1.71 (1.33-2.19)	1.49 (1.16-1.92)	1.36 (1.02-1.80)
Large LV	21/3134	6.7 (4.4-10.3)	2.6 (-0.3-5.5)	1.64 (1.06-2.55)	1.48 (0.95-2.30)	1.31 (0.78-2.21)
Women						

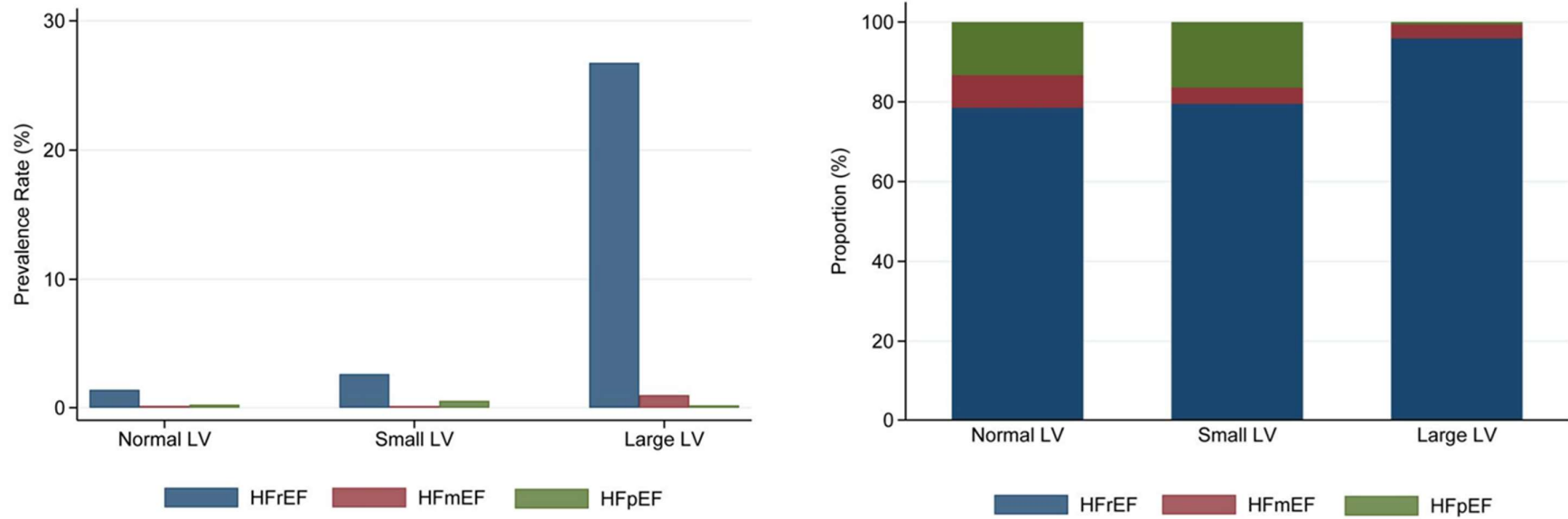
MACE						
Normal LV	454/95163	4.8 (4.4-5.2)	Ref.	Ref.	Ref.	Ref.
Small LV	69/11730	5.9 (4.6-7.4)	1.1 (-0.3-2.6)	1.25 (0.97-1.61)	1.07 (0.83-1.38)	1.01 (0.75-1.36)
Large LV	17/2676	6.4 (3.9-10.2)	1.6 (-1.5-4.6)	1.33 (0.82-2.16)	1.30 (0.80-2.12)	1.16 (0.64-2.12)
Coronary heart disease						
Normal LV	304/96499	3.2 (2.8-3.5)	Ref.	Ref.	Ref.	Ref.
Small LV	52/11977	4.3 (3.3-5.7)	1.2 (0-2.4)	1.40 (1.04-1.88)	1.17 (0.87-1.58)	1.10 (0.82-1.48)
Large LV	8/2733	2.9 (1.5-5.9)	-0.2 (-2.3-1.8)	0.93 (0.46-1.87)	0.91 (0.45-1.84)	0.98 (0.50-1.90)
Heart failure						
Normal LV	98/105661	0.69 (0.8-1.1)	Ref.	Ref.	Ref.	Ref.
Small LV	12/13429	0.9 (0.5-1.6)	0 (-0.6-0.5)	0.97 (0.53-1.77)	0.77 (0.42-1.41)	0.85 (0.43-1.66)
Large LV	16/3033	5.3 (3.2-8.6)	4.3 (1.8-6.9)	5.79 (3.41-9.82)	5.67 (3.32-9.68)	6.14 (3.30-11.44)
Ischemic stroke						
Normal LV	96/105665	0.9 (0.7-1.1)	Ref.	Ref.	Ref.	Ref.
Small LV	17/13422	1.3 (0.8-2.0)	0.4 (-0.3-1.0)	1.40 (0.84-2.34)	1.22 (0.72-2.06)	1.30 (0.72-2.35)
Large LV	3/3110	1.0 (0.3-3.0)	0 (-1.1-1.2)	1.06 (0.34-3.34)	0.95 (0.30-3.02)	0.89 (0.22-3.66)
Death from any cause						
Normal LV	210/101183	2.1 (1.8-2.4)	Ref.	Ref.	Ref.	Ref.
Small LV	34/12877	2.6 (1.9-3.7)	0.6 (-0.4-1.5)	1.30 (0.91-1.87)	1.19 (0.82-1.71)	1.05 (0.68-1.63)
Large LV	6/2980	2.0 (0.9-4.5)	0 (-1.7-1.6)	0.96 (0.43-2.16)	0.95 (0.42-2.14)	0.86 (0.32-2.32)

Model 1: Adjusted for age, sex and race.

Model 2: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index and total cholesterol.

Model 3: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and LVEF.

Figure S1. The prevalence of heart failure subtypes stratified by LV size.



LV = left ventricle; HFrEF = heart failure with reduced ejection fraction; HFmEF = heart failure with moderate reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction.

Figure S2. Incidence rate of small LV and large LV and change as a function of age.

